Clinical Risk Assessment Report: Patient 1

1. Primary Outcome & Risk Summary

- Target: Occult Lymph Node (OLN) Metastasis
- Model Prediction: Negative for Metastasis
- Prediction Confidence: Borderline / Low Confidence
- Absolute Predicted Risk: 19.5% (Diagnostic Threshold: 22.1%)
- Relative Risk Multiplier: 0.88x the diagnostic threshold

Clinical Takeaway:

The machine learning model classified this patient as Low-Risk for Occult Lymph Node (OLN) Metastasis. The primary driver determining this prediction is the Small Zone Emphasis (PET) (GLZLM_SZE.PET), which reduced the patient's absolute risk by 6.1%. Biologically, this feature reflects the proportion of small, fine texture zones, often correlating with a highly chaotic and erratic micro-architecture.

Clinical Sensitivity Analysis (Borderline Patient):

PREDICTION INSTABILITY WARNING: This patient's risk profile sits precisely on the borderline. Consequently, the model's prediction is highly sensitive. Minor biological variations or measurement errors in the following features would flip the final clinical prediction:

- Small Zone Emphasis (PET): An increase would alter the prediction outcome.
- 75th Percentile Tumor Density: A decrease would alter the prediction outcome.
- Squamous Cell Carcinoma Antigen: A decrease would alter the prediction outcome.

2. Key Pathological & Imaging Drivers (Elevating Risk)

Large Zone Emphasis (CT) (GLZLM_LGZE.CT) **Value: -0.44 [Bottom <1%] (+4.7%)**

Reflects the presence of large homogeneous areas; lower values suggest a chaotic, heterogeneous microenvironment.

Short Run High Gray-Level Emphasis (CT) (GLRLM_SRHGE.CT) **Value: 1.12 [Top 8%] (+3.3%)**

Reflects fragmented, short streaks of high density or metabolism, indicating a disjointed and aggressive tumor architecture.

Patient Age (Age) **Value: 56.0 [Bottom 25%] (+1.0%)**

Patient age influences immune surveillance, tissue microenvironment, and baseline metabolic rates.

High Gray-Level Zone Emphasis (CT) (GLZLM_HGZE.CT) **Value: 1.15 [Top 10%] (+0.8%)**

Highlights the distribution of high-density or highly metabolic active regions clustered within the tumor.

Tumor Sphericity (PET) (SHAPE_Sphericity(onlyFor3DROI)).PET) **Value: 0.49 [Avg (66th %ile)] (+0.7%)**

Quantifies how closely the tumor resembles a perfect sphere; lower values indicate irregular, spiculated, or highly invasive margins.

3. Protective / Mitigating Factors (Reducing Risk)

Small Zone Emphasis (PET) (GLZLM_SZE.PET) **Value: -1.13 [Bottom 13%] (-6.1%)**

Reflects the proportion of small, fine texture zones, often correlating with a highly chaotic and erratic micro-architecture.

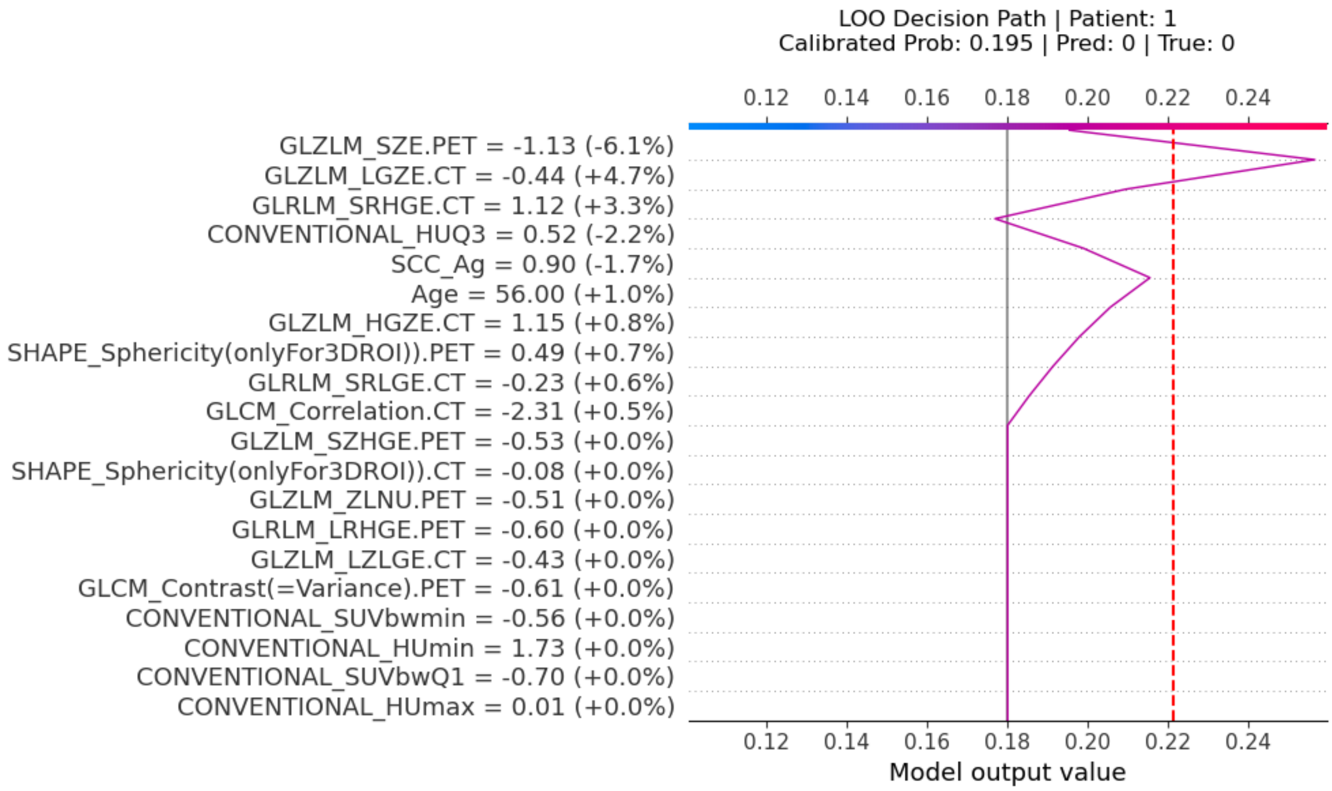
75th Percentile Tumor Density (CONVENTIONAL_HUQ3) **Value: 0.52 [Avg (60th %ile)] (-2.2%)**

Reflects the overall physical density, solid components, and general attenuation characteristics of the tumor.

Squamous Cell Carcinoma Antigen (SCC_Ag) **Value: 0.9 [Avg (74th %ile)] (-1.7%)**

Serum levels reflect tumor proliferation, burden, and squamous cell differentiation.

Machine Learning Feature Attribution (SHAP Decision Path)



Sensitivity Analysis: Feature Tipping Points

The plots below demonstrate how variations in specific clinical or radiomic features affect the model's predicted risk. The red marker indicates the patient's current clinical state. Crossing the red dashed line changes the diagnosis.